

# Muhammed Usman

## GenAI Developer

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### Professional Summary

Generative AI Python Developer with 2+ years of experience building and deploying production-ready LLM applications. Proficient in Python, LLM, LangChain, RAG pipelines, and Azure cloud services. Worked on a real-time AI system exploring prompt engineering, vector databases (PG vector, FAISS) and Flask API for robust backend services. Passionate about scalable AI engineering, DevOps, and building impactful user experiences using LLM frameworks.

### Technical Skills

**Languages:** Python, SQL

**Frameworks & Libraries:** LangChain, Flask, Pandas, NumPy, GenAI & NLP: GPT-4, Prompt Engineering, RAG, FAISS, DevOps & Cloud: Microsoft Azure, Azure DevOps, Blob Storage

**Testing:** Manual Testing and Regression Testing

### Work Experience

#### Tata Consultancy Services (TCS) – Python GenAI Developer

Nov 2023 – Present

- Worked on full-stack GenAI chatbots using GPT-4, LangChain, and Azure services for enterprise-level document-based Q&A.
- Developed prompt templates and RAG pipelines using PG vector search to support semantic search across company documents and used prompt engineering techniques to adapt model's output.
- Designed Python-based APIs with Flask for real-time LLM inference and data retrieval.
- Integrated Azure Blob Storage, Key Vault, and DevOps pipelines for secure, scalable deployments across dev and prod environments.
- Applied LangChain tools to orchestrate modular tasks within LLM pipelines.

### Education

**M.Sc. in Computer Science (Data Science)** | Cochin University of Science and Technology (CUSAT), 2021 – 2023

**Bachelor of Computer Applications (B.C.A.)** | Kannur University, 2018 – 2021

### Projects

- **LOA Chatbot** – Created a document-based GPT-4 chatbot with vector search, Azure integration, and document retrieval using RAG technique and Prompt Engineering. Explored

Langchain features like chat history retrieval, stuff document, etc.

- **Internal SOP Chatbot** – Developed a GPT-based chatbot for marine operations support, integrating LangChain and RAG for document retrieval, with backend on Flask and vector indexing over internal SOPs.
- **Heart Failure Data Visualization** – Built a Flask dashboard using Python, Pandas, and NumPy to visualize patient data trends.
- **Real-Time Speech Recognition with AR** – Implemented a CNN-RNN model for speech-to-text and integrated with AR interface for accessibility.

## Languages

English, Hindi, Malayalam

## Certifications

Microsoft Azure DP-100 Data Scientist Associate